Math 32 Curriculum

Precalculus: Trigonometry | Middlesex School Mathematics | Resource: Math 32 Workbook

COURSE AT A GLANCE

Trigonometric Foundations

REQUIRED

pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions; pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 13-14 Arc Length; pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions; pp. 22-23 Right Triangle Applications

UNIT LEARNING OBJECTIVES

I can use the Pythagorean theorem to determine an unknown side of a right triangle.

I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.

I can convert angle measures between degrees and radians.

I can determine coterminal and reference angles for a given angle.

I can calculate arc length from a central angle and radius.

I can determine exact coordinates on the unit circle.

I can evaluate all six trigonometric functions exactly for common angles.

I can solve right triangles using trigonometric ratios.

I can apply right-triangle trigonometry to contextual problems.

I can evaluate trigonometric functions from a point on a circle or terminal side.

I can solve a basic trigonometric equation for all angles in a specified interval.

Periodic Functions and Modeling

REQUIRED

pp. 46-49 Graphing Sine and Cosine; pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations; pp. 69-76 Modeling Problems; pp. 77-85 Graphing Cosecant, Secant, and Tangent

UNIT LEARNING OBJECTIVES

I can graph basic sine and cosine functions using key features.

I can identify amplitude, midline, period, and phase shift from an equation or graph.

I can graph transformed sine and cosine functions.

I can write sine and cosine equations that represent a periodic graph.

I can construct a sinusoidal model from contextual information or data.

I can interpret the parameters and outputs of a sinusoidal model in context.

I can graph tangent, secant, and cosecant functions with their asymptotes.

Inverse Functions and Trigonometric Equations

REQUIRED

pp. 86-91 Calculating Angles; Inverse Trig Functions; pp. 92-98 Solving Trig Equations, Part 2; pp. 99-100 Compositions of Trig and Inverse Trig Functions

UNIT LEARNING OBJECTIVES

I can read inverse-trigonometric notation and distinguish it from reciprocal notation.

I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.

I can explain how restricted domains make inverse trigonometric functions possible.

I can solve trigonometric equations exactly or approximately on a specified interval.

I can express all solutions of a trigonometric equation in general form.

I can evaluate compositions of trigonometric and inverse trigonometric functions.

Non-Right Triangles

REQUIRED

pp. 102-110 Law of Sines; Ambiguous Case; pp. 111-115 Law of Cosines; Mixed Practice

UNIT LEARNING OBJECTIVES

I can solve a non-right triangle using the Law of Sines.

I can determine the number of possible triangles in the ambiguous SSA case.

I can solve a non-right triangle using the Law of Cosines.

I can select and apply an appropriate method to solve a triangle in context.

Trigonometric Identities

REQUIRED

pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

UNIT LEARNING OBJECTIVES

I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.

I can rewrite trigonometric expressions using the Pythagorean identities.

I can use even/odd and cofunction identities to rewrite trigonometric expressions.

I can verify a trigonometric identity using the required identity families.

Additional Trigonometric Identities

EXTENSION

pp. 117-118, 124-126 Double-angle and sum/difference identities

UNIT LEARNING OBJECTIVES

I can apply double-angle identities.

I can apply half-angle identities.

I can apply angle sum and difference identities.

I can determine exact trigonometric values using extension identities.

Triangle Area from SAS

EXTENSION

pp. 101 New Area Formula

UNIT LEARNING OBJECTIVES

I can calculate the area of a triangle from two sides and the included angle.

Polar Coordinates

REQUIRED

pp. 127-128 Polar Coordinates and Coordinate Conversion

UNIT LEARNING OBJECTIVES

I can plot and identify points in polar coordinates.

I can represent a polar point using equivalent coordinate pairs.

I can convert points between polar and rectangular coordinates.

Polar Functions

EXTENSION

pp. 129-132 Polar Functions; pp. 133-135 Polar/Rectangular Function Conversion

UNIT LEARNING OBJECTIVES

I can graph and interpret basic polar functions.

I can convert equations between polar and rectangular forms.

Vectors

EXTENSION

pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

UNIT LEARNING OBJECTIVES

I can represent a vector geometrically and in component form.

I can add, subtract, and multiply vectors by scalars.

I can determine vector magnitude, direction, and unit vector.

I can resolve a vector into horizontal and vertical components.

I can perform basic operations with vectors in three dimensions.

Parametric Equations

EXTENSION

pp. 145-147 Introduction to Parametrics

UNIT LEARNING OBJECTIVES

I can graph a plane curve defined by parametric equations with its orientation.

I can eliminate a parameter to obtain a rectangular equation.

I can create or interpret parametric equations for motion in the plane.

IN-DEPTH CURRICULUM

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Trigonometric Foundations

REQUIRED

Resource coverage: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions; pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 13-14 Arc Length; pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions; pp. 22-23 Right Triangle Applications

M32-FND-001

I can use the Pythagorean theorem to determine an unknown side of a right triangle.

Resource: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- R Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN · first introduced in Math 22

M32-FND-002

I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.

Resource: pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Distinguish exact values from measured or rounded approximations.
  SK-GEO-EXACT-APPROX · first introduced in Math 22
- A Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- R Recall special-right-triangle ratios.
  SK-TRI-SPECIAL-RATIOS · first introduced in Math 22

M32-FND-003

I can convert angle measures between degrees and radians.

Resource: pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Add, subtract, multiply, and divide fractions.
  SK-NUM-FRACTION-OPERATE · inherited from middle school
- I Convert between degree and radian measure.
  SK-TRIG-ANGLE-CONVERT

M32-FND-004

I can determine coterminal and reference angles for a given angle.

Resource: pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 29-33 Review Questions

SUPPORTING SKILLS

- I Generate and recognize coterminal angles.
  SK-TRIG-ANGLE-COTERMINAL
- I Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE
- I Determine the sign of a trigonometric function by quadrant.
  SK-TRIG-SIGN-QUADRANT

M32-FND-005

I can calculate arc length from a central angle and radius.

Resource: pp. 13-14 Arc Length; pp. 29-33 Review Questions

SUPPORTING SKILLS

- D Calculate arc length from radius and central angle.
  SK-CIR-ARC-LENGTH · first introduced in Math 22
- A Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22
- A Convert between degree and radian measure.
  SK-TRIG-ANGLE-CONVERT · first introduced in Math 32

M32-FND-006

I can determine exact coordinates on the unit circle.

Resource: pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Recall special-right-triangle ratios.
  SK-TRI-SPECIAL-RATIOS · first introduced in Math 22
- A Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- I Determine unit-circle coordinates for common angles.
  SK-TRIG-UNIT-CIRCLE-COORD

M32-FND-007

I can evaluate all six trigonometric functions exactly for common angles.

Resource: pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 29-33 Review Questions; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions

SUPPORTING SKILLS

- I — Express common trigonometric values exactly.
  SK-TRIG-EXACT-VALUE
- I — Recall the six trigonometric functions, their reciprocal relationships, and their notation.
  SK-TRIG-NOTATION-VOCAB
- I — Relate sine/cosecant, cosine/secant, and tangent/cotangent.
  SK-TRIG-RECIPROCAL
- A — Determine the sign of a trigonometric function by quadrant.
  SK-TRIG-SIGN-QUADRANT · first introduced in Math 32
- A — Determine unit-circle coordinates for common angles.
  SK-TRIG-UNIT-CIRCLE-COORD · first introduced in Math 32

M32-FND-008

I can solve right triangles using trigonometric ratios.

Resource: pp. 22-23 Right Triangle Applications; pp. 29-33 Review Questions

SUPPORTING SKILLS

- D — Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- D — Use an inverse trigonometric function to determine a missing angle.
  SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- D — Use a trigonometric ratio to determine a missing side.
  SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22
- R — Label sides relative to a selected angle.
  SK-TRIG-TRIANGLE-LABEL · first introduced in Math 22

M32-FND-009

I can apply right-triangle trigonometry to contextual problems.

Resource: pp. 22-23 Right Triangle Applications; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A — Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A — Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12
- D — Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- A — Use an inverse trigonometric function to determine a missing angle.
  SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- A — Use a trigonometric ratio to determine a missing side.
  SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22

M32-FND-010

I can evaluate trigonometric functions from a point on a circle or terminal side.

Resource: pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 29-33 Review Questions

SUPPORTING SKILLS

- A Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN · first introduced in Math 22
- I Use coordinates and radius to evaluate trigonometric functions.
  SK-TRIG-CIRCLE-POINT
- A Relate sine/cosecant, cosine/secant, and tangent/cotangent.
  SK-TRIG-RECIPROCAL · first introduced in Math 32
- A Determine the sign of a trigonometric function by quadrant.
  SK-TRIG-SIGN-QUADRANT · first introduced in Math 32

M32-FND-011

I can solve a basic trigonometric equation for all angles in a specified interval.

Resource: pp. 34-45 Multiple Angle Solutions; Reciprocal Functions

SUPPORTING SKILLS

- A Translate between inequality and interval notation.
  SK-NOT-INTERVAL · first introduced in Math 12
- A Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- I Use reference angles and periodicity to find every solution in an interval.
  SK-TRIG-EQUATION-INTERVAL
- I Isolate a trigonometric expression before solving.
  SK-TRIG-EQUATION-ISOLATE
- A Determine the sign of a trigonometric function by quadrant.
  SK-TRIG-SIGN-QUADRANT · first introduced in Math 32

Periodic Functions and Modeling

REQUIRED

Resource coverage: pp. 46-49 Graphing Sine and Cosine; pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations; pp. 69-76 Modeling Problems; pp. 77-85 Graphing Cosecant, Secant, and Tangent

M32-GRF-001

I can graph basic sine and cosine functions using key features.

Resource: pp. 46-49 Graphing Sine and Cosine

SUPPORTING SKILLS

- D Use tabular patterns to predict graph shape and behavior.
  SK-REP-TABLE-GRAPH · first introduced in Math 21
- I Locate quarter-period key points of a periodic function.
  SK-TRIG-GRAPH-KEYPOINTS
- A Determine unit-circle coordinates for common angles.
  SK-TRIG-UNIT-CIRCLE-COORD · first introduced in Math 32

M32-GRF-002

I can identify amplitude, midline, period, and phase shift from an equation or graph.

Resource: pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- I Determine amplitude and vertical reflection.
  SK-TRIG-GRAPH-AMPLITUDE
- I Determine a sinusoid's midline and vertical shift.
  SK-TRIG-GRAPH-MIDLINE
- I Determine period from a graph or equation.
  SK-TRIG-GRAPH-PERIOD
- I Determine horizontal shift from a graph or equation.
  SK-TRIG-GRAPH-PHASE
- I Recall and correctly use the terms amplitude, midline, period, phase shift, vertical shift, and asymptote.
  SK-TRIG-GRAPH-VOCAB

M32-GRF-003

I can graph transformed sine and cosine functions.

Resource: pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- A Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- D Distinguish stretches from compressions and horizontal effects from vertical effects.
  SK-FUNC-TRANSFORM-SCALE · first introduced in Math 31
- A Interpret transformations applied to function outputs.
  SK-FUNC-TRANSFORM-VERTICAL · first introduced in Math 31
- D Locate quarter-period key points of a periodic function.
  SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32

M32-GRF-004

I can write sine and cosine equations that represent a periodic graph.

Resource: pp. 63-68 Graphing All Transformations

SUPPORTING SKILLS

- D Connect parameters in a formula to features of its graph.
  SK-REP-FORMULA-GRAPH · first introduced in Math 21
- D Determine amplitude and vertical reflection.
  SK-TRIG-GRAPH-AMPLITUDE · first introduced in Math 32
- D Determine a sinusoid's midline and vertical shift.
  SK-TRIG-GRAPH-MIDLINE · first introduced in Math 32
- D Determine period from a graph or equation.
  SK-TRIG-GRAPH-PERIOD · first introduced in Math 32
- D Determine horizontal shift from a graph or equation.
  SK-TRIG-GRAPH-PHASE · first introduced in Math 32

M32-GRF-005

I can construct a sinusoidal model from contextual information or data.

Resource: pp. 69-76 Modeling Problems

SUPPORTING SKILLS

- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- A Determine amplitude and vertical reflection.
  SK-TRIG-GRAPH-AMPLITUDE · first introduced in Math 32
- A Determine a sinusoid's midline and vertical shift.
  SK-TRIG-GRAPH-MIDLINE · first introduced in Math 32
- A Determine period from a graph or equation.
  SK-TRIG-GRAPH-PERIOD · first introduced in Math 32
- I Translate contextual maximum, minimum, cycle, and starting position into model parameters.
  SK-TRIG-MODEL-PARAMETERS

M32-GRF-006

I can interpret the parameters and outputs of a sinusoidal model in context.

Resource: pp. 69-76 Modeling Problems

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12
- D Translate contextual maximum, minimum, cycle, and starting position into model parameters.
  SK-TRIG-MODEL-PARAMETERS · first introduced in Math 32

M32-GRF-007

I can graph tangent, secant, and cosecant functions with their asymptotes.

Resource: pp. 77-85 Graphing Cosecant, Secant, and Tangent

SUPPORTING SKILLS

- A Interpret transformations applied to function inputs.
  SK-FUNC-TRANSFORM-HORIZONTAL · first introduced in Math 31
- I Locate vertical asymptotes of tangent, secant, and cosecant.
  SK-TRIG-GRAPH-ASYMPTOTE
- A Locate quarter-period key points of a periodic function.
  SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32
- D Relate sine/cosecant, cosine/secant, and tangent/cotangent.
  SK-TRIG-RECIPROCAL · first introduced in Math 32

Inverse Functions and Trigonometric Equations REQUIRED

Resource coverage: pp. 86-91 Calculating Angles; Inverse Trig Functions; pp. 92-98 Solving Trig Equations, Part 2; pp. 99-100 Compositions of Trig and Inverse Trig Functions

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M32-INV-006

I can read inverse-trigonometric notation and distinguish it from reciprocal notation.

Resource: No mapped textbook section

SUPPORTING SKILLS

- I — Read inverse-trigonometric notation and distinguish it from reciprocal notation.
  SK-TRIG-INVERSE-NOTATION
- A — Relate sine/cosecant, cosine/secant, and tangent/cotangent.
  SK-TRIG-RECIPROCAL · first introduced in Math 32

M32-INV-001

I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.

Resource: pp. 86-91 Calculating Angles; Inverse Trig Functions

SUPPORTING SKILLS

- A — Express common trigonometric values exactly.
  SK-TRIG-EXACT-VALUE · first introduced in Math 32
- I — Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE
- A — Read inverse-trigonometric notation and distinguish it from reciprocal notation.
  SK-TRIG-INVERSE-NOTATION · first introduced in Math 32
- I — Recall the principal-value ranges of inverse trigonometric functions.
  SK-TRIG-INVERSE-RANGE

M32-INV-002

I can explain how restricted domains make inverse trigonometric functions possible.

Resource: pp. 86-91 Calculating Angles; Inverse Trig Functions

SUPPORTING SKILLS

- A — Exchange domain and range between a function and its inverse.
  SK-FUNC-INVERSE-DOMAIN-RANGE · first introduced in Math 31
- A — Apply the horizontal line test for invertibility.
  SK-FUNC-INVERSE-HLT · first introduced in Math 31
- A — Test whether a function is one-to-one on a domain.
  SK-FUNC-ONE-TO-ONE · first introduced in Math 31
- D — Recall the principal-value ranges of inverse trigonometric functions.
  SK-TRIG-INVERSE-RANGE · first introduced in Math 32

M32-INV-003

I can solve trigonometric equations exactly or approximately on a specified interval.

Resource: pp. 92-98 Solving Trig Equations, Part 2

SUPPORTING SKILLS

- A Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- A Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32
- D Use reference angles and periodicity to find every solution in an interval.
  SK-TRIG-EQUATION-INTERVAL · first introduced in Math 32
- D Isolate a trigonometric expression before solving.
  SK-TRIG-EQUATION-ISOLATE · first introduced in Math 32
- A Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32

M32-INV-004

I can express all solutions of a trigonometric equation in general form.

Resource: pp. 92-98 Solving Trig Equations, Part 2

SUPPORTING SKILLS

- D Generate and recognize coterminal angles.
  SK-TRIG-ANGLE-COTERMINAL · first introduced in Math 32
- I Express a periodic solution family in general form.
  SK-TRIG-EQUATION-GENERAL
- A Use reference angles and periodicity to find every solution in an interval.
  SK-TRIG-EQUATION-INTERVAL · first introduced in Math 32

M32-INV-005

I can evaluate compositions of trigonometric and inverse trigonometric functions.

Resource: pp. 99-100 Compositions of Trig and Inverse Trig Functions

SUPPORTING SKILLS

- A Substitute one function formula into another.
  SK-FUNC-COMPOSE-FORMULA · first introduced in Math 31
- A Verify inverse functions using composition.
  SK-FUNC-INVERSE-VERIFY · first introduced in Math 31
- D Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- A Recall the principal-value ranges of inverse trigonometric functions.
  SK-TRIG-INVERSE-RANGE · first introduced in Math 32

Non-Right Triangles REQUIRED

Resource coverage: pp. 102-110 Law of Sines; Ambiguous Case; pp. 111-115 Law of Cosines; Mixed Practice

M32-TRI-002

I can solve a non-right triangle using the Law of Sines.

Resource: pp. 102-110 Law of Sines; Ambiguous Case

SUPPORTING SKILLS

- A Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- I Apply the Law of Sines to determine missing sides or angles.
  SK-TRIG-LAW-SINES
- D Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

M32-TRI-003

I can determine the number of possible triangles in the ambiguous SSA case.

Resource: pp. 102-110 Law of Sines; Ambiguous Case

SUPPORTING SKILLS

- I Test and interpret the possible outcomes of SSA data.
  SK-TRIG-AMBIGUOUS-CASE
- D Apply the Law of Sines to determine missing sides or angles.
  SK-TRIG-LAW-SINES · first introduced in Math 32
- A Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

M32-TRI-004

I can solve a non-right triangle using the Law of Cosines.

Resource: pp. 111-115 Law of Cosines; Mixed Practice

SUPPORTING SKILLS

- A Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32
- I Apply the Law of Cosines to determine missing sides or angles.
  SK-TRIG-LAW-COSINES
- D Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA · first introduced in Math 32

M32-TRI-005

I can select and apply an appropriate method to solve a triangle in context.

Resource: pp. 111-115 Law of Cosines; Mixed Practice

SUPPORTING SKILLS

- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12
- A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- A Apply the Law of Cosines to determine missing sides or angles.
  SK-TRIG-LAW-COSINES · first introduced in Math 32
- A Apply the Law of Sines to determine missing sides or angles.
  SK-TRIG-LAW-SINES · first introduced in Math 32
- D Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA · first introduced in Math 32
- I Select right-triangle ratios, Law of Sines, or Law of Cosines from the given data.
  SK-TRIG-TRIANGLE-METHOD

Trigonometric Identities REQUIRED

Resource coverage: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

M32-IDN-001

I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- I — Apply reciprocal and quotient identities.
  SK-TRIG-IDENTITY-RECIPROCAL-QUOTIENT
- I — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE

M32-IDN-002

I can rewrite trigonometric expressions using the Pythagorean identities.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- I — Apply the Pythagorean trigonometric identities.
  SK-TRIG-IDENTITY-PYTHAGOREAN
- D — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDN-003

I can use even/odd and cofunction identities to rewrite trigonometric expressions.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- D — Test a formula or graph for even or odd symmetry.
  SK-FUNC-SYMMETRY · first introduced in Math 31
- I — Apply sine/cosine cofunction identities.
  SK-TRIG-IDENTITY-COFUNCTION
- I — Apply even and odd trigonometric identities.
  SK-TRIG-IDENTITY-EVEN-ODD
- D — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDN-004

I can verify a trigonometric identity using the required identity families.

Resource: pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

SUPPORTING SKILLS

- A — Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I — Choose a productive identity or algebraic manipulation.
  SK-TRIG-IDENTITY-STRATEGY
- D — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32
- I — Maintain equivalence while transforming one side of an identity.
  SK-TRIG-IDENTITY-VERIFY

Additional Trigonometric Identities

EXTENSION

Resource coverage: pp. 117-118, 124-126 Double-angle and sum/difference identities

M32-IDX-001

I can apply double-angle identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

- I — Apply double-angle identities.
  SK-TRIG-IDENTITY-DOUBLE-ANGLE
- A — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32

M32-IDX-002

I can apply half-angle identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

- A — Apply double-angle identities.
  SK-TRIG-IDENTITY-DOUBLE-ANGLE · first introduced in Math 32
- I — Apply half-angle identities.
  SK-TRIG-IDENTITY-HALF-ANGLE

M32-IDX-003

I can apply angle sum and difference identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

- A — Substitute an identity to rewrite an expression.
  SK-TRIG-IDENTITY-SUBSTITUTE · first introduced in Math 32
- I — Apply angle sum and difference identities.
  SK-TRIG-IDENTITY-SUM-DIFFERENCE

M32-IDX-004

I can determine exact trigonometric values using extension identities.

Resource: pp. 117-118, 124-126 Double-angle and sum/difference identities

SUPPORTING SKILLS

- D — Express common trigonometric values exactly.
  SK-TRIG-EXACT-VALUE · first introduced in Math 32
- A — Apply double-angle identities.
  SK-TRIG-IDENTITY-DOUBLE-ANGLE · first introduced in Math 32
- A — Apply half-angle identities.
  SK-TRIG-IDENTITY-HALF-ANGLE · first introduced in Math 32
- A — Apply angle sum and difference identities.
  SK-TRIG-IDENTITY-SUM-DIFFERENCE · first introduced in Math 32

Triangle Area from SAS

EXTENSION

Resource coverage: pp. 101 New Area Formula

M32-TRX-001

I can calculate the area of a triangle from two sides and the included angle.

Resource: pp. 101 New Area Formula

SUPPORTING SKILLS

- A Express common trigonometric values exactly.
  SK-TRIG-EXACT-VALUE · first introduced in Math 32
- I Calculate triangle area from two sides and the included angle.
  SK-TRIG-TRIANGLE-AREA-SAS
- I Classify triangle data as SSS, SAS, ASA, AAS, or SSA.
  SK-TRIG-TRIANGLE-DATA

Polar Coordinates REQUIRED

Resource coverage: pp. 127-128 Polar Coordinates and Coordinate Conversion

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M32-POL-001

I can plot and identify points in polar coordinates.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I Read and write polar-coordinate notation and identify the radius, angle, pole, and polar axis.
  SK-POL-NOTATION-READ
- I Plot a point from radius and angle.
  SK-POL-PLOT
- A Determine a reference angle and its quadrant.
  SK-TRIG-ANGLE-REFERENCE · first introduced in Math 32

M32-POL-002

I can represent a polar point using equivalent coordinate pairs.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I Generate equivalent polar coordinate pairs.
  SK-POL-EQUIVALENT
- A Generate and recognize coterminal angles.
  SK-TRIG-ANGLE-COTERMINAL · first introduced in Math 32

M32-POL-003

I can convert points between polar and rectangular coordinates.

Resource: pp. 127-128 Polar Coordinates and Coordinate Conversion

SUPPORTING SKILLS

- I Use coordinate relationships to convert a point between systems.
  SK-POL-CONVERT-POINT
- A Use coordinates and radius to evaluate trigonometric functions.
  SK-TRIG-CIRCLE-POINT · first introduced in Math 32
- A Evaluate an inverse trigonometric expression exactly or approximately.
  SK-TRIG-INVERSE-EVALUATE · first introduced in Math 32

Polar Functions EXTENSION

Resource coverage: pp. 129-132 Polar Functions; pp. 133-135 Polar/Rectangular Function Conversion

M32-PLX-001

I can graph and interpret basic polar functions.

Resource: pp. 129-132 Polar Functions

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- D Plot a point from radius and angle.
  SK-POL-PLOT · first introduced in Math 32
- A Locate quarter-period key points of a periodic function.
  SK-TRIG-GRAPH-KEYPOINTS · first introduced in Math 32

M32-PLX-002

I can convert equations between polar and rectangular forms.

Resource: pp. 133-135 Polar/Rectangular Function Conversion

SUPPORTING SKILLS

- A Simplify an algebraic expression using operation properties.
  SK-ALG-EXPRESSION-SIMPLIFY · inherited from middle school
- I Substitute coordinate identities to convert an equation between systems.
  SK-POL-CONVERT-EQUATION
- A Apply the Pythagorean trigonometric identities.
  SK-TRIG-IDENTITY-PYTHAGOREAN · first introduced in Math 32

Vectors EXTENSION

Resource coverage: pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

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M32-VEC-001

I can represent a vector geometrically and in component form.

Resource: pp. 136-140 Vector Concepts and Operations

SUPPORTING SKILLS

- A Add congruence, parallel, perpendicular, and angle markings to a diagram.
  SK-GEO-DIAGRAM-MARK · first introduced in Math 22
- I Write a vector in component form.
  SK-VEC-COMPONENTS
- I Read and write vector notation and correctly use the terms magnitude, direction, component, and unit vector.
  SK-VEC-NOTATION-VOCAB

M32-VEC-002

I can add, subtract, and multiply vectors by scalars.

Resource: pp. 136-140 Vector Concepts and Operations

SUPPORTING SKILLS

- A Add, subtract, multiply, and divide signed integers.
  SK-NUM-INTEGER-OPERATE · inherited from middle school
- I Perform component-wise vector operations.
  SK-VEC-OPERATE

M32-VEC-003

I can determine vector magnitude, direction, and unit vector.

Resource: pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- A Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
  SK-COORD-DISTANCE-PYTHAGOREAN · first introduced in Math 22
- I Determine a vector's direction angle with correct quadrant.
  SK-VEC-DIRECTION
- I Calculate vector magnitude using the distance formula.
  SK-VEC-MAGNITUDE
- I Normalize a nonzero vector to obtain a unit vector.
  SK-VEC-UNIT

M32-VEC-004

I can resolve a vector into horizontal and vertical components.

Resource: pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- D Write a vector in component form.
  SK-VEC-COMPONENTS · first introduced in Math 32
- A Determine a vector's direction angle with correct quadrant.
  SK-VEC-DIRECTION · first introduced in Math 32

M32-VEC-005

I can perform basic operations with vectors in three dimensions.

Resource: pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

SUPPORTING SKILLS

- I Represent and operate on vectors with three components.
  SK-VEC-3D
- D Calculate vector magnitude using the distance formula.
  SK-VEC-MAGNITUDE · first introduced in Math 32
- D Perform component-wise vector operations.
  SK-VEC-OPERATE · first introduced in Math 32

Parametric Equations

EXTENSION

Resource coverage: pp. 145-147 Introduction to Parametrics

M32-PAR-001

I can graph a plane curve defined by parametric equations with its orientation.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A Identify input and output values in a formula, graph, table, or context.
  SK-FUNC-INPUT-OUTPUT · first introduced in Math 12
- I Graph a parametric curve with its direction of motion.
  SK-PAR-GRAPH
- I Read parametric notation and correctly use the terms parameter, coordinate equation, orientation, and parameter interval.
  SK-PAR-NOTATION-VOCAB
- I Generate ordered pairs from parametric equations.
  SK-PAR-TABLE

M32-PAR-002

I can eliminate a parameter to obtain a rectangular equation.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A Isolate a specified variable in an equation or formula.
  SK-ALG-VARIABLE-ISOLATE · first introduced in Math 12
- I Eliminate a parameter algebraically.
  SK-PAR-ELIMINATE
- A Substitute an equivalent expression for a variable.
  SK-SYS-SUBSTITUTE · first introduced in Math 12

M32-PAR-003

I can create or interpret parametric equations for motion in the plane.

Resource: pp. 145-147 Introduction to Parametrics

SUPPORTING SKILLS

- A Explain a model parameter in the language of its context.
  SK-MODEL-PARAMETER-INTERPRET · first introduced in Math 21
- I Interpret or construct parametric equations for planar motion.
  SK-PAR-MODEL
- D Generate ordered pairs from parametric equations.
  SK-PAR-TABLE · first introduced in Math 32
- A Attach and interpret units for rates, inputs, and outputs.
  SK-REP-UNITS · first introduced in Math 12